

AI Resume Screening Agent

Project Report

Abstract

The AI Resume Screening Agent automates resume screening using Artificial Intelligence and Natural Language Processing. It compares resumes with job descriptions, extracts skills, experience, and education, calculates a matching score, and recommends suitable candidates. This reduces manual effort and improves recruitment efficiency.

Problem Statement

Manual resume screening is time-consuming and may overlook qualified candidates. An AI-based solution can automate the process and improve consistency.

Objectives

- Automate resume screening
- Compare resumes with job descriptions
- Rank candidates
- Generate AI recommendations
- Reduce hiring time

Tools and Technologies

Python, Flask, HTML, CSS, JavaScript, SQLite/MySQL, OpenAI/Gemini API, GitHub, VS Code.

Modules

1. Login Module
2. Resume Upload
3. Resume Parsing
4. Job Description Analysis
5. AI Matching
6. Candidate Ranking
7. Report Generation

Algorithm

1. Start
2. Upload Job Description
3. Upload Resume
4. Extract Resume Text

5. Analyze Skills
6. Compare with Job Description
7. Calculate Match Score
8. Rank Candidates
9. Generate Recommendation
10. End

Advantages

Fast screening, reduced manual work, improved accuracy, consistent evaluation.

Limitations

Depends on resume quality and AI model accuracy.

Future Scope

Support multiple languages, video interview analysis, LinkedIn integration, cloud deployment.

Conclusion

The AI Resume Screening Agent improves recruitment by automating resume analysis and helping recruiters identify suitable candidates quickly and efficiently.